Unit V: Plant Physiology (Functional Organisation)

Chapter 13

Photosynthesis



Learning Objectives

The learner will be able to,

- *Learn the Ultra structure of Chloroplast .*
- *Realise the importance of solar energy and properties of light.*
- *Acquire knowledge of Quantum, Quantum yield and Quantum requirement.*
- *Develop curiosity for photosynthetic experiments like Red drop, Emerson Enhancement effect and Hill's Reaction.*
- *Analyse the pathway of electron- PS I and PS II.*
- *Recognise the Photo-Oxidative and Photo Chemical Pathway.*
- *Develop skill in Photosynthetic pathways and ability to draw C_3, C_4, C_2 and CAM cycle.*

Chapter Outline

13.1 Definition, Significance and Site of photosynthesis
13.2 Photosynthetic pigments
13.3 Spectrum of electromagnetic radiation
13.4 Photosynthetic unit
13.5 Absorption spectrum and Action spectrum
13.6 Emerson's experiments & Hill's reaction
13.7 Modern concept of photosynthesis
13.8 Photo-oxidation phase of light reaction
13.9 Photochemical phase of light reaction
13.10 Photophosphorylation
13.11 Dark reaction or C_3 cycle
13.12 Hatch & Slack Pathway or C_4 Cycle
13.13 CAM cycle or Crassulacean Acid Metabolism
13.14 Photorespiration or C_2 Cycle
13.15 Factors affecting photosynthesis
13.16 Photosynthesis in bacteria

Life on earth is made up of organic compounds. How do we get these organic compounds? Ultimately, plants are the main source of all kinds of carbon compounds in this planet. We directly or indirectly depend on plants for this. Plants are the major machinery which produce organic compounds like carbohydrates, lipids, proteins, nucleic acids and other biomolecules.

Though man has reached the glory of achievements still he is not able to imitate the metabolic activities of plants which produces energy resources and other biomolecules.

The plants get energy from sun by converting solar or radiant energy into chemical energy by the process of Photosynthesis, which acts as a driving force for both biotic and abiotic world. Photosynthesis produces 1700 million tonnes of dry matter per year by fixing 75×10^{12} Kg of carbon every year. Photosynthetic organisms use only 0.2 % of incident solar light on earth. Carbohydrates produced by photosynthesis are the basic raw material for respiration and also to produce many organic compounds. It maintains atmospheric oxygen and carbon dioxide level. Photosynthesis consumes atmospheric carbon dioxide which is continuously added by the respiration of organisms. Photosynthesis is the

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